DATA SCIENCE
&
ARTIFICIAL INTELLIGENCE



Calculus

Lecture No. 01



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Topics to be Covered



Topic

FUNCTIONS & THEIR GRAPHS

Function :- f is said to be a func^n from A to B if

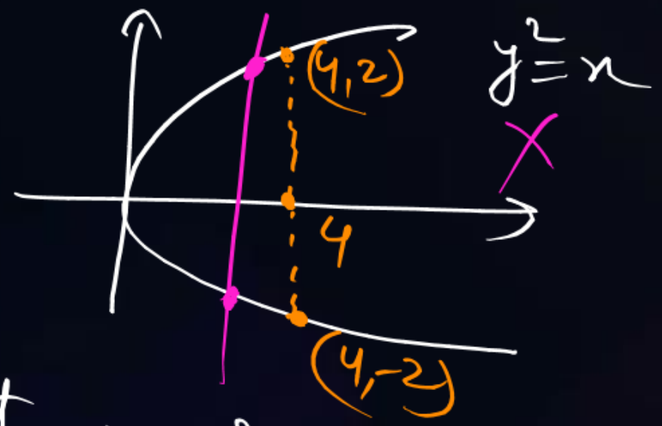
$\forall\ x \in A$ $\exists$ unique $y \in B$ s.t $y = f(x)$

Here x = Independent Variable

y = Dependent Variable.

Domain ↓ Codomain

for eg $y^2 = x$ is not a func^n bcoz At $x = 4$, $y = +2$ or $y = -2$



ie uniqueness property does not Hold.

But $y = x^2$ is a func^n



Trick: if line $\parallel^r$ to y axis cuts the graph only at one point then it is func^n.

Domain :- Set of permissible values of x for which func^n is defined is called Domain

Range :- " " " " " of y " " " " " " Range

Note :- Range is either equal to Codomain or a subset of Codomain

ie $\boxed{\text{Range } f \subseteq \text{Codomain}}$ → Either Range = Codomain
or Range ⊂ Codomain

Periodic func^n :- if Graph of func^n repeats after certain value of x then it is called Periodic

ie $f(x)$ is said to be a periodic func^n of period T if $\boxed{f(x+T) = f(x)}$

for eg. $\sin x$ is periodic func^n of period 2π $\because \sin(x+2\pi) = \sin x$

" $\tan x$ " " " " " π " $\because \tan(x+\pi) = \tan x$

Even func^n → if Graph of func^n is symmetrical about Y axis then it is called Even fun^n.

i.e if $f(-x) = f(x)$ then $f(x)$ is called Even func^n.

odd func^n → if Graph of func^n is symmetrical about origin then it is called odd func^n

i.e it must be symmetrical in opposite Quadrant, I $\longleftrightarrow$ III, II $\longleftrightarrow$ IV

if $f(-x) = -f(x)$ then $f(x)$ is called odd func^n

Neither Even Nor odd → if Graph of func^n is neither symmetrical about Y axis Nor symmetrical about origin then it is called NENO func^n.

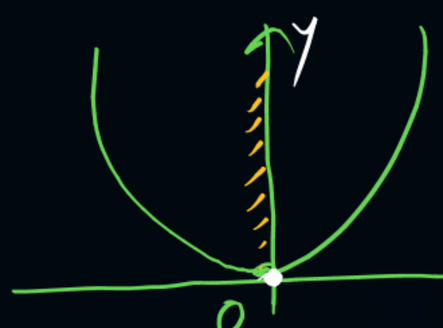
if $f(-x) \neq f(x)$ $\neq -f(x)$ then $f(x)$ is called NENO func^n

Types of func^n

① Algebraic func^n
- ① Polynomial func^n
- ② Rational func^n
- ③ Irrational func^n
- ④ Piecewise func^n
  - ① Modulus func^n
  - ② Signum func^n
  - ③ Greatest Integer func^n
  - ④ Fractional Part func^n

② Transcendental func^n
- ① Trignometric func^n
- ② Inverse Trig func^n
- ③ Exponential func^n
- ④ log func^n.

① $y = x^2, x^4, x^6 \ldots\ldots$



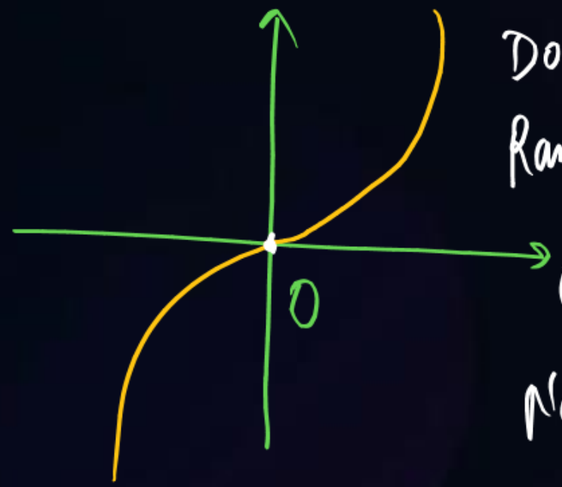
Dom $= (-\infty, \infty)$

Range $= [0, \infty)$

Even func^n

Non Periodic

② $y = x^3, x^5, x^7, \ldots\ldots$



Dom $= (-\infty, \infty)$

Range $= (-\infty, \infty)$

odd func^n

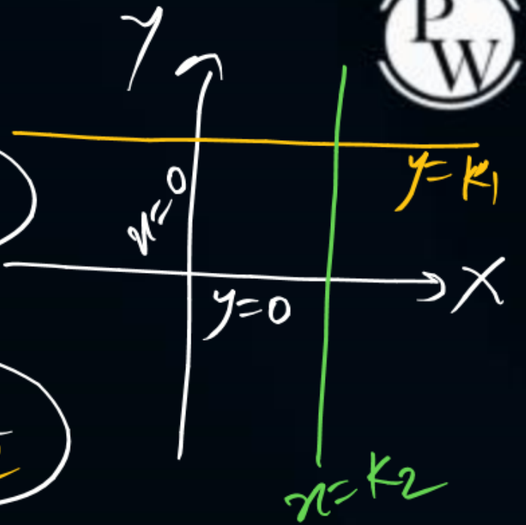
Non periodic

③ Equ^n of x axis is $y = 0$

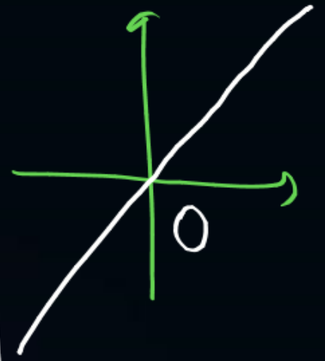
& equ^n of line $\parallel^r$ to x axis is $y = k_1$

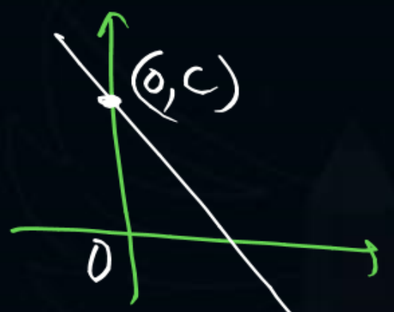
Equ^n of y axis is $x = 0$

& equ^n of line $\parallel^r$ to y axis is $x = k_2$

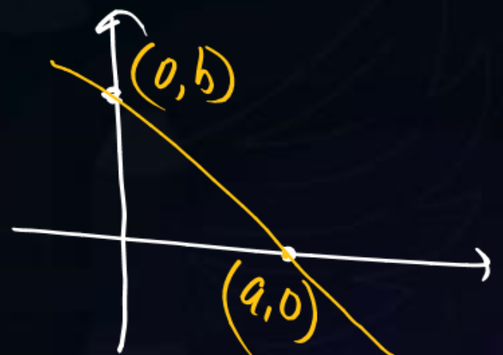


④ $y = mx$; $y = mx + c$; $\frac{x}{a} + \frac{y}{b} = 1$



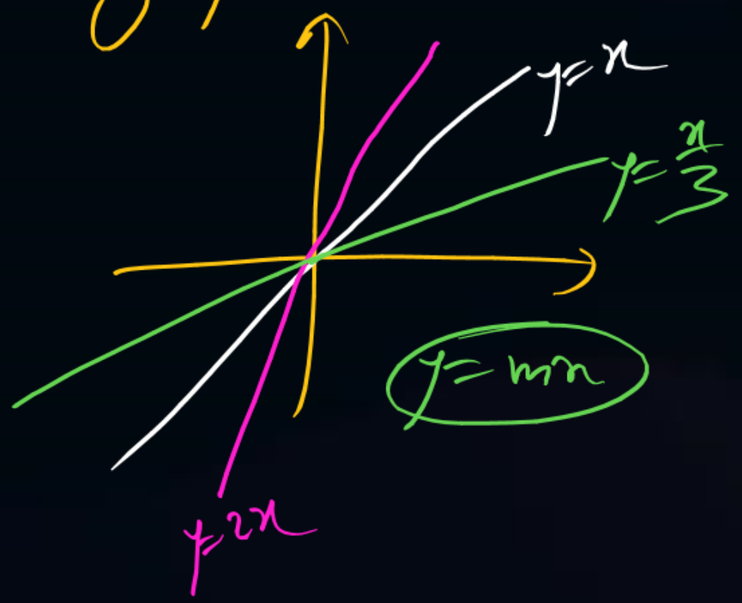


(Slope Intercept form)



Intercept form

e.g. $y = x$, $y = 2x$, $y = 3x$ .....



e.g. $y = mx + c$

$-mx + y = c$

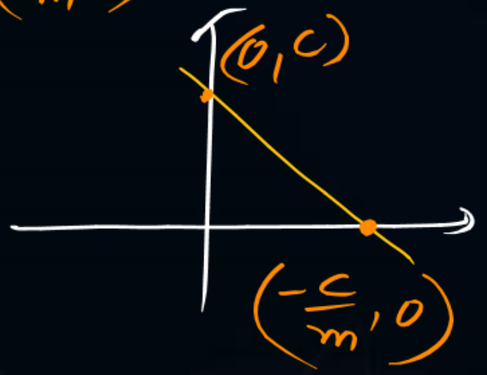
$\frac{-mx}{c} + \frac{y}{c} = 1$

$$\boxed{\frac{x}{(-c/m)} + \frac{y}{c} = 1}$$

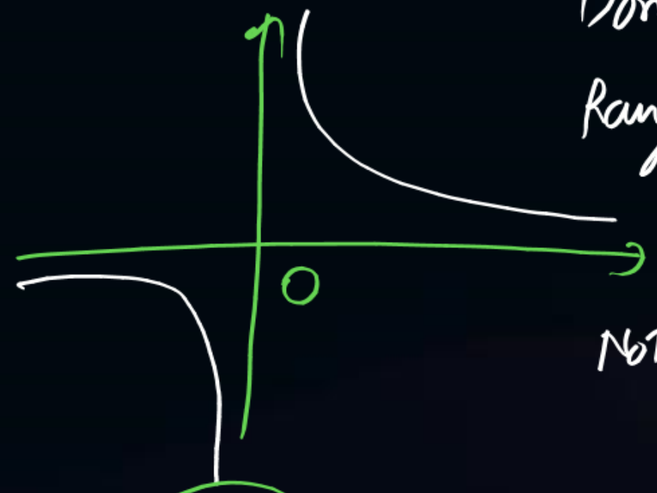
if m = +ve



if m = -ve



(5) $y=\frac{1}{x}$, $\frac{1}{x^3}$, $\frac{1}{x^5}$, ------



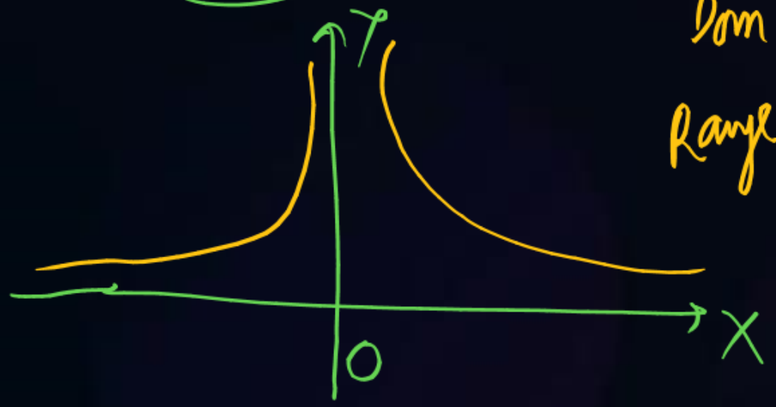
Dom = $R-\{0\}$

Range = $R-\{0\}$

odd func^n

Not periodic

(6) $y=\frac{1}{x^2}$, $\frac{1}{x^4}$, $\frac{1}{x^6}$ ------



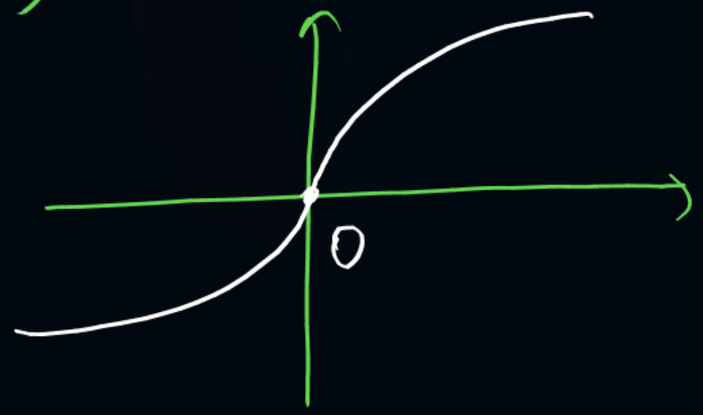
Dom = $R-\{0\}$

Range = $(0,\infty)$

Even func^n

Not Periodic

(7) $y=x^{1/3}$, $x^{1/5}$, $x^{1/7}$ ------



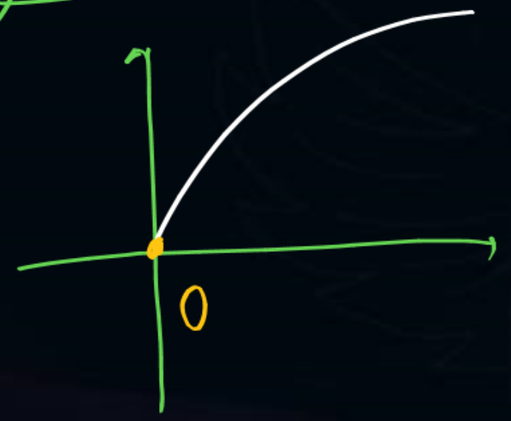
Dom = $(-\infty,\infty)$

Range = $(-\infty,\infty)$

odd func^n

Non Periodic

(8) $y=x^{1/2}$, $x^{1/4}$, $x^{1/6}$ ------



Dom = $[0,\infty)$

Range = $[0,\infty)$

NENO.

Non Periodic

Explanation :

① For Discrete x : if $1 \le x \le 5 \Rightarrow x = 1, 2, 3, 4, 5$

if $1 < x < 5 \Rightarrow x = 2, 3, 4$

② for Continuous x, if $1 \le x \le 5 \Rightarrow x \in [1, 5] \Rightarrow x = 1, 1.1, 1.01, 1.5 \ldots 4.9, 5$

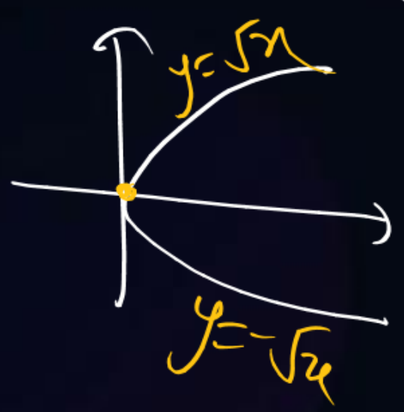
if $1 < x < 5 \Rightarrow x \in (1, 5) \Rightarrow x = 1.01, 1.1, 1.2 \ldots 4.8, 4.9$

$x \neq 1$ & $x \neq 5$

③ $y^2 = x \Rightarrow y = \pm\sqrt{x}$

$y = +\sqrt{x}$

$y = -\sqrt{x}$





(9) $y = e^x$



Dom $= (-\infty, \infty)$

Range $= (0, \infty)$

NEND

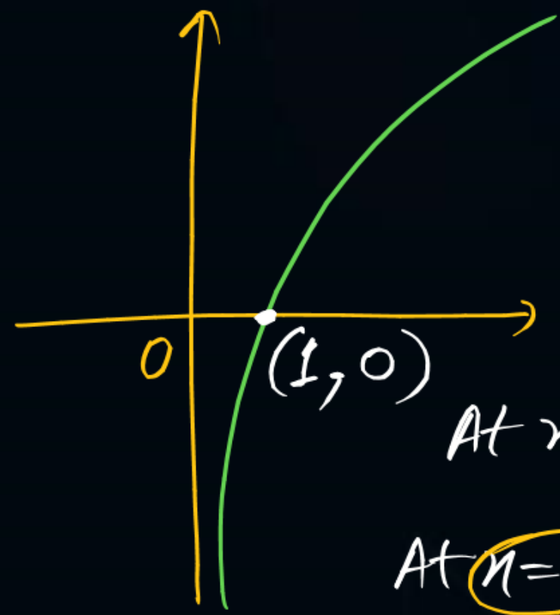
Non Periodic

At $x = 0$, $y(0) = e^0 = 1$

At $x = -\infty$, $y(-\infty) = e^{-\infty} = 0$ Lower

At $x = +\infty$, $y(+\infty) = e^{\infty} = +\infty$ "

(10) $y = \log_e x = \ln x$



Dom $= (0, \infty)$

Range $= (-\infty, \infty)$

NEND

Non Periodic

At $x = 1$, $y(1) = \log_e 1 = 0$

At $x = 0$, $y(0) = \log_e(0) = -\infty$ ∉ ND

At $x = +\infty$, $y(\infty) = \log_e(\infty) = +\infty$ ∉ ND

i.e $\log x$ is defined only for true values of x i.e for $x > 0$

for eg $\log(-3) =$ N.D

Some More Properties of log →

① $\log(ab) = \log a + \log b$

② $\log\left(\frac{a}{b}\right) = \log a - \log b$

③ $\log(a^b) = b \log a$

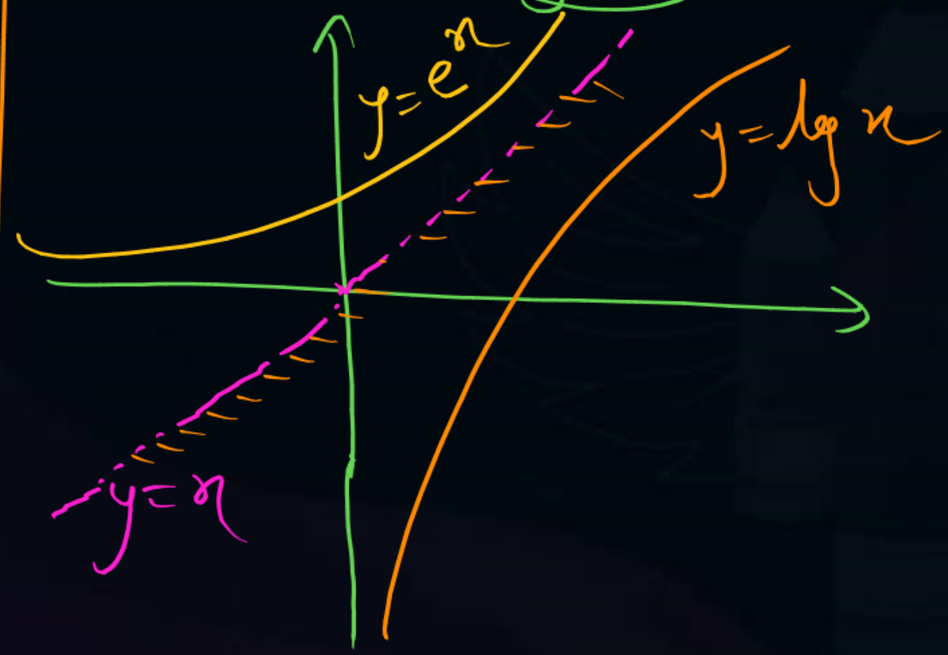
But $(\log a)^b \neq b \log a$

④ $\log_b a = \frac{\log_e a}{\log_e b}$, $b \neq 1$

eg $\log_1 a = \frac{\log_e a}{\log_e 1} = \frac{\log_e a}{0} = N.D$

⑤ $\log_b a = \frac{1}{\log_a b}$; ⑥ $\log_{a^n} x = \frac{1}{n} \log_a x$

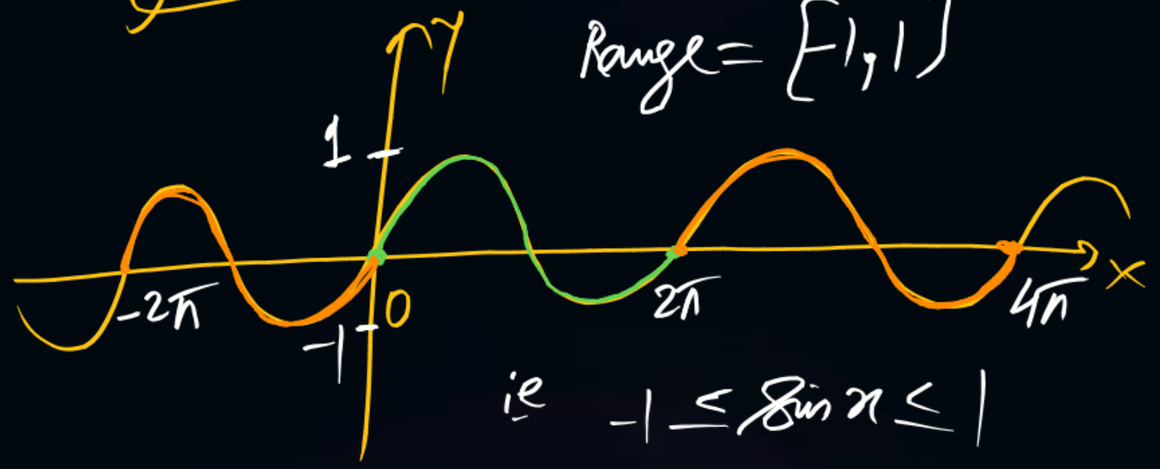
⑦ $y = e^x$ & $y = \log_e x$ are Inverse func^n of each other bcoz they can be obtained by taking Mirror image about the line $y = x$



(11) $y = \sin x$

$\text{Dom} = (-\infty, \infty)$

$\text{Range} = [-1, 1]$



ie $-1 \le \sin x \le 1$

odd func^n & Periodic with period $= 2\pi$

(12) $y = \cos x$

$\text{Dom} = (-\infty, \infty)$

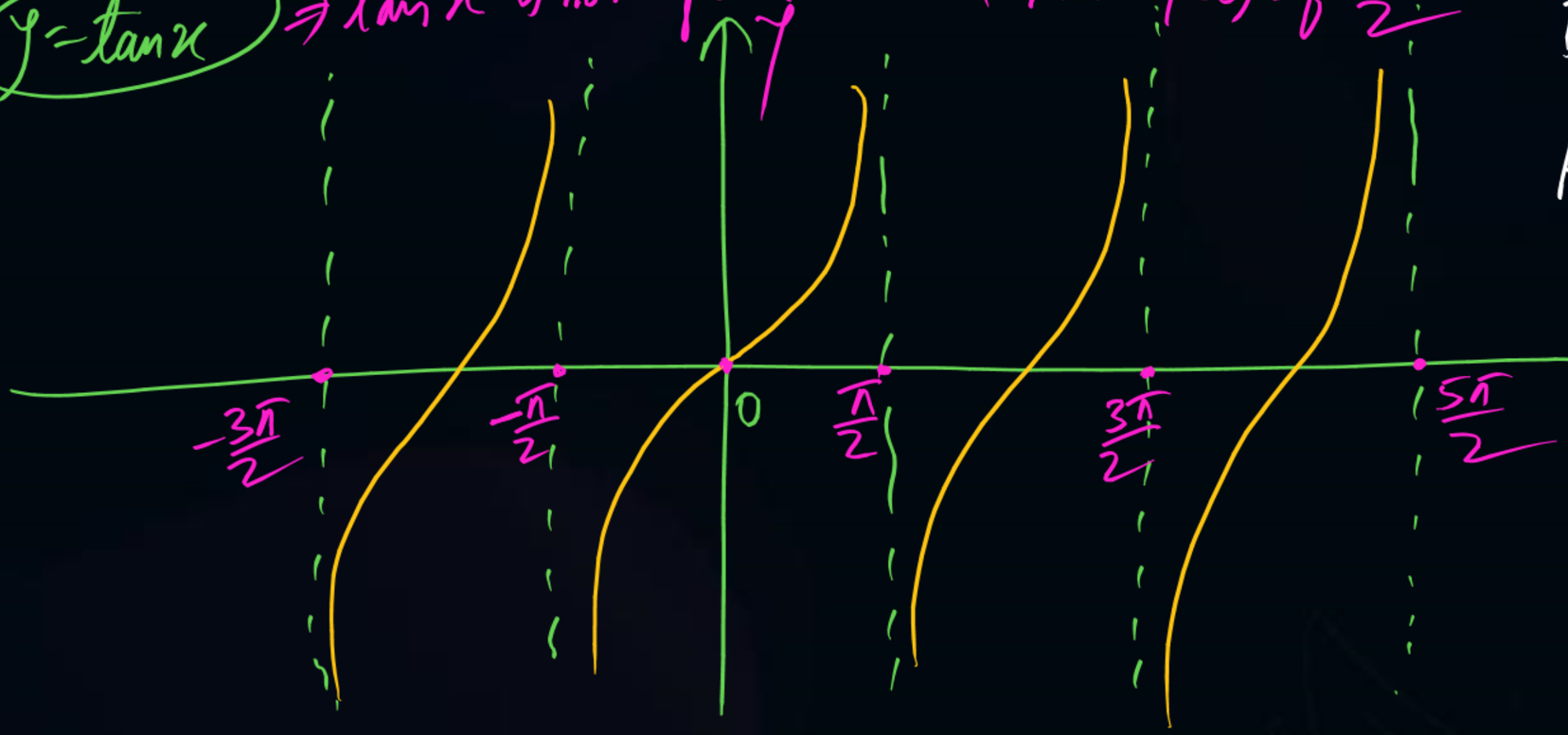
$\text{Range} = [-1, 1]$



ie $-1 \le \cos x \le 1$

Even func^n; Periodic with period $= 2\pi$

(B) $y = \tan x$ $\Rightarrow$ $\tan x$ is not defined at odd Multiples of $\frac{\pi}{2}$.



Dom $= R - (2n+1)\frac{\pi}{2}$

Range $= (-\infty, \infty)$

odd func^n,

Periodic with period $= \pi$

Note: To Represent Integers, we should take $= n$ i.e $n = \ldots -3, -2, -1, 0, 1, 2, 3 \ldots$

" " Even Integers, " " $= 2n$ i.e $n = \ldots -4, -2, 0, 2, 4, 6 \ldots$

" " odd " " " " $= (2n+1)$ i.e $n = \ldots -3, -1, 1, 3, 5, \ldots$

Q. one of the sol. of $\tan x = x$ can be approximated as?

(a) 1.57

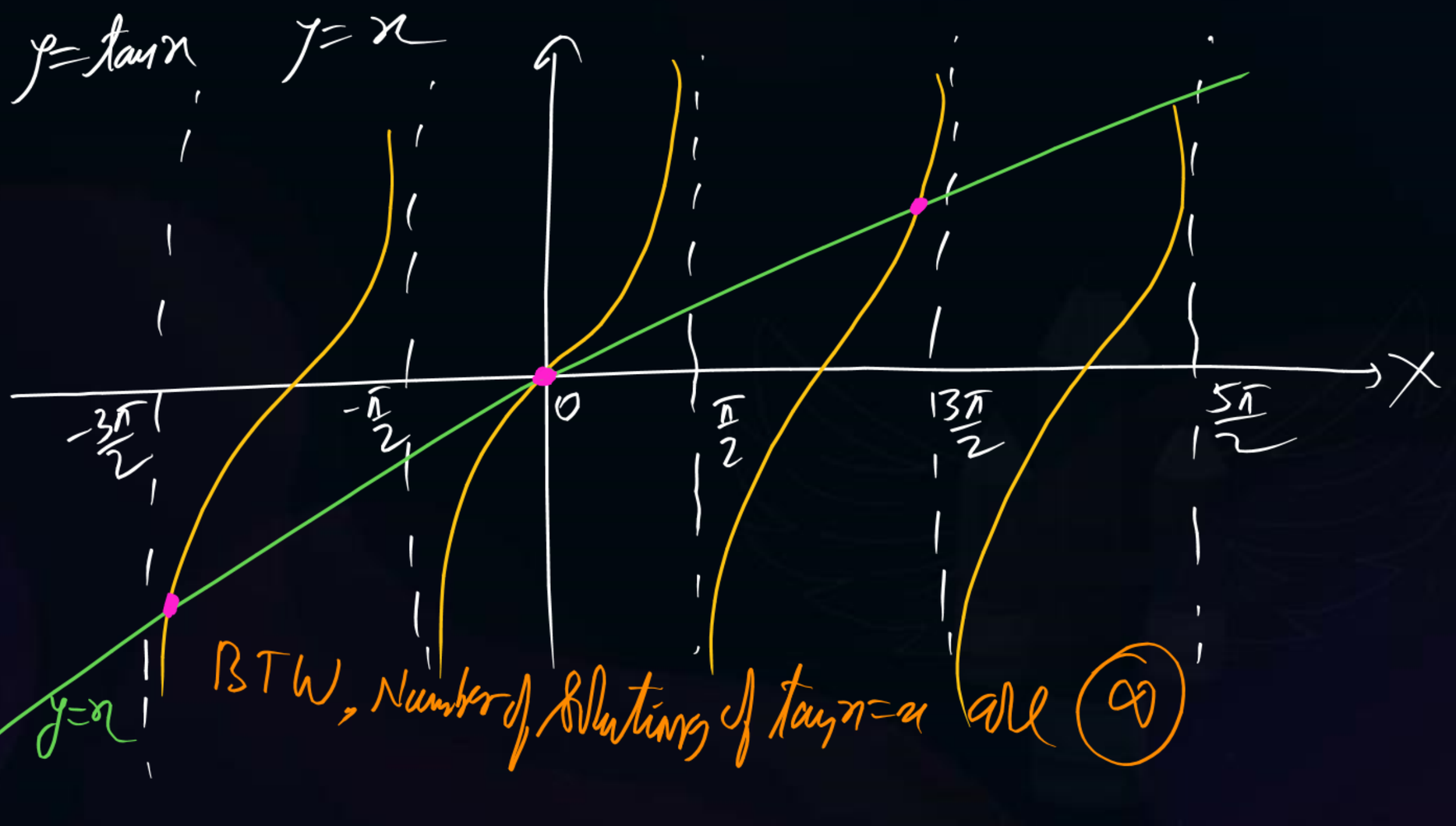
(b) 3.14

(c) 4.5

(d) None

$x \simeq \frac{3\pi}{2} = \frac{3 \times 3.14}{2}$

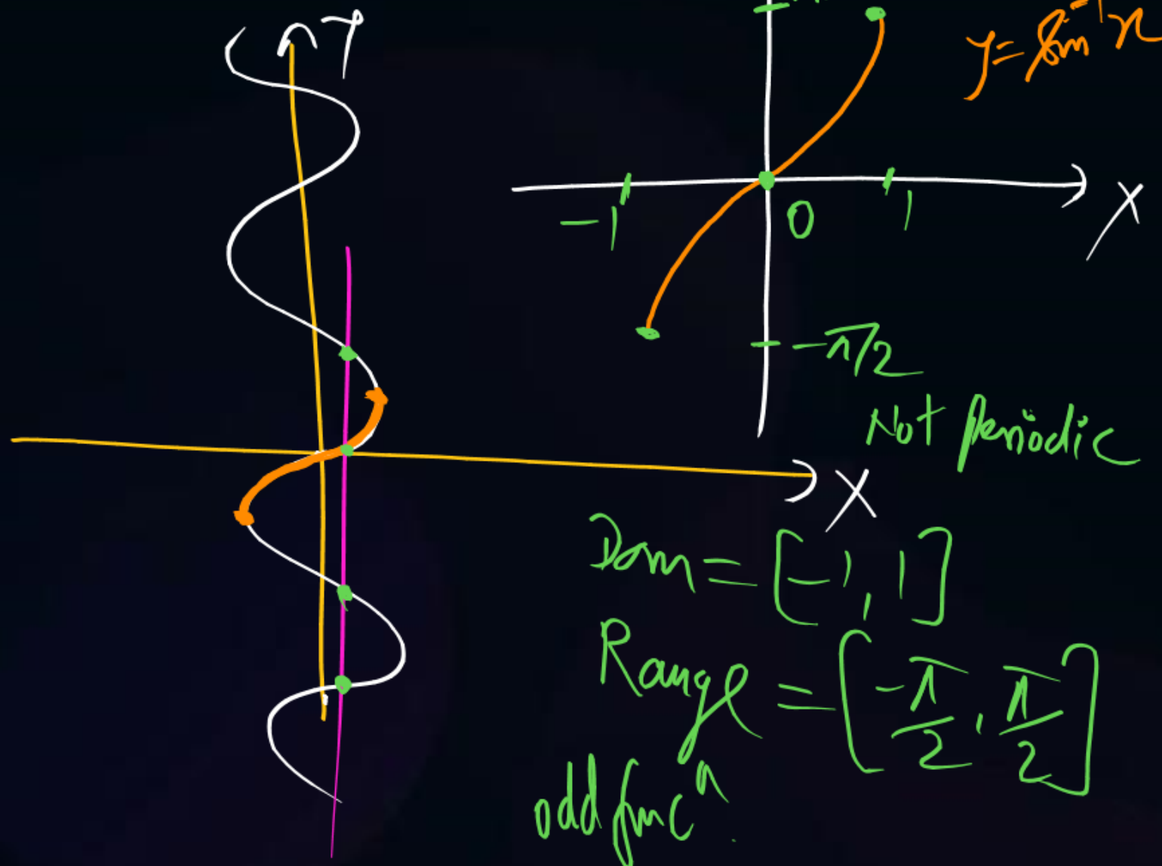
$\simeq 4.5$



BTW, Number of solutions of $\tan x = x$ are ∞

(14) $y = \sin^{-1} x$





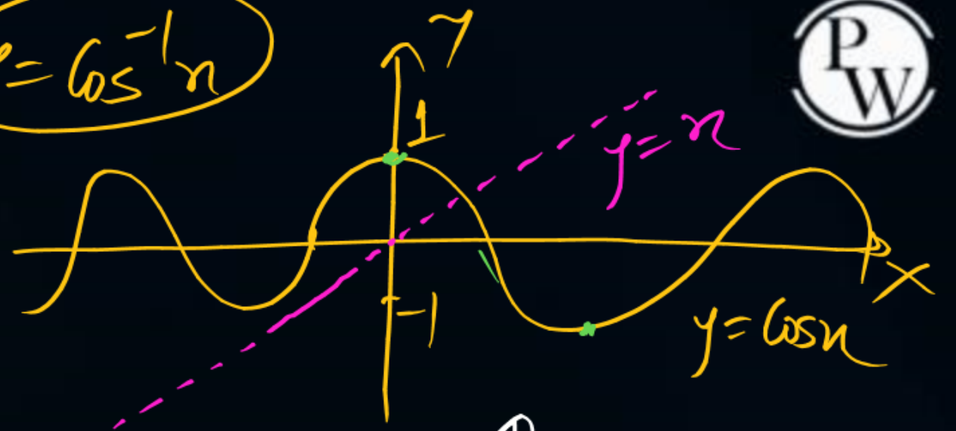
Not Periodic

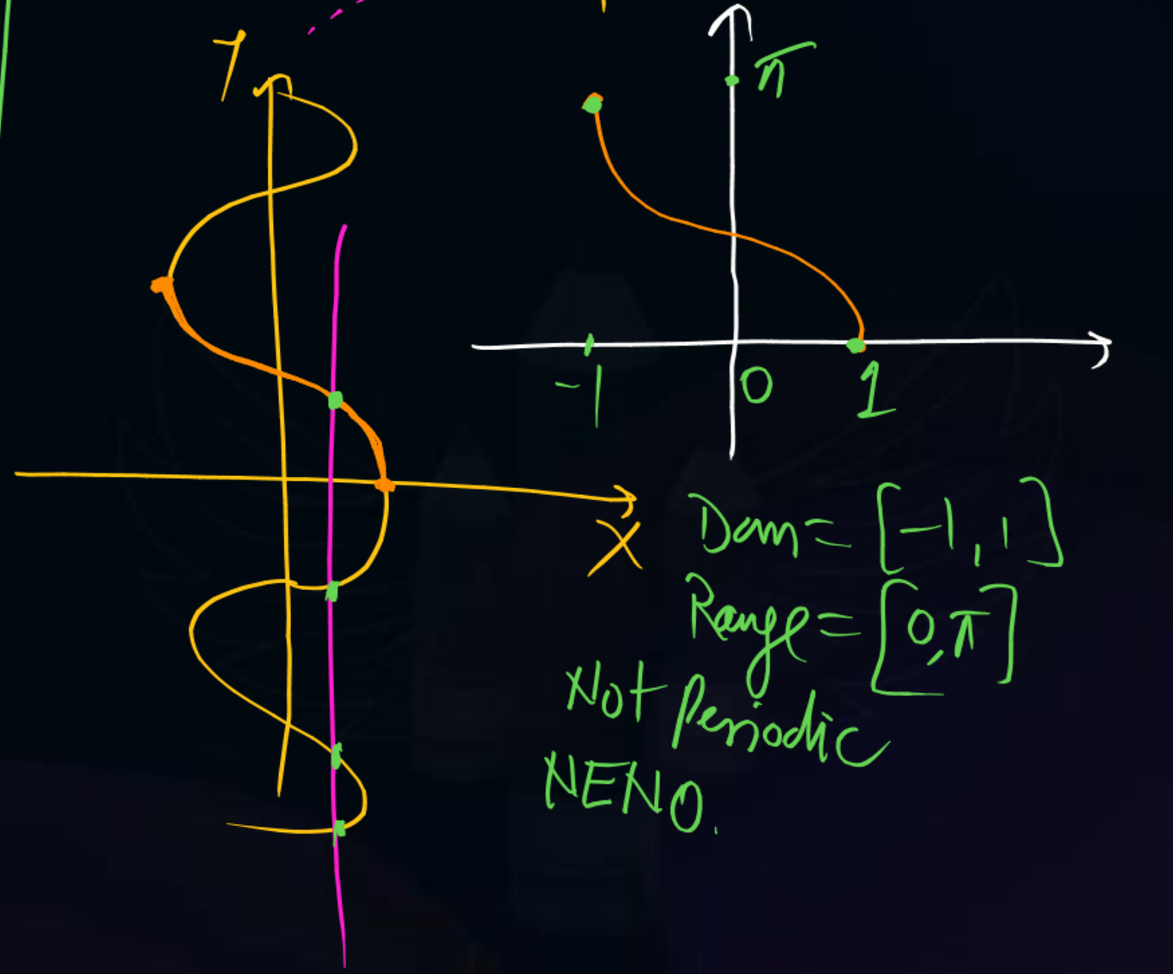
Dom = $[-1, 1]$

Range = $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

odd func^{n}.

(15) $y = \cos^{-1} x$





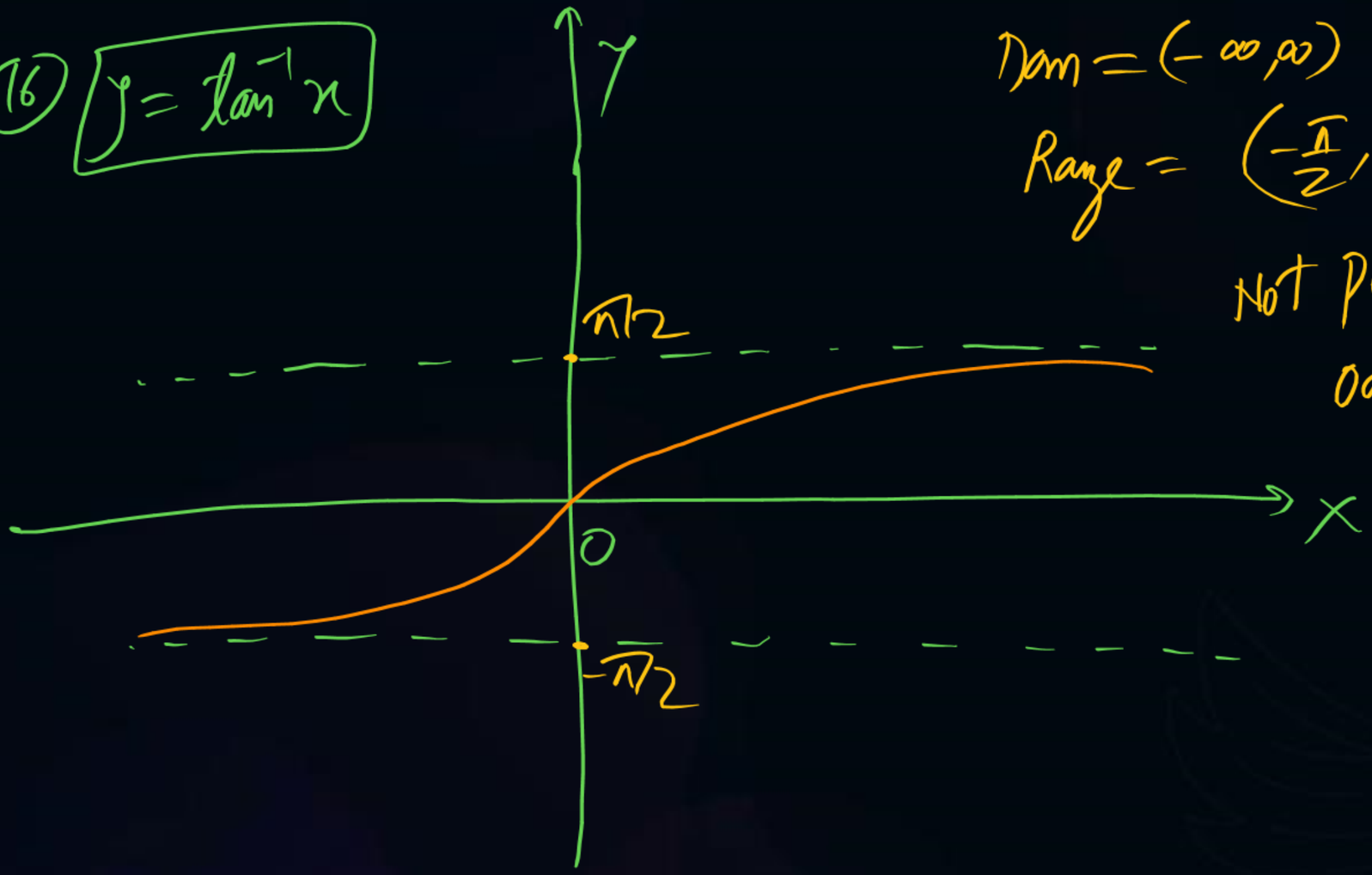
Dom = $[-1, 1]$

Range = $[0, \pi]$

Not Periodic

NEITHER.

(16) $y = \tan^{-1} x$



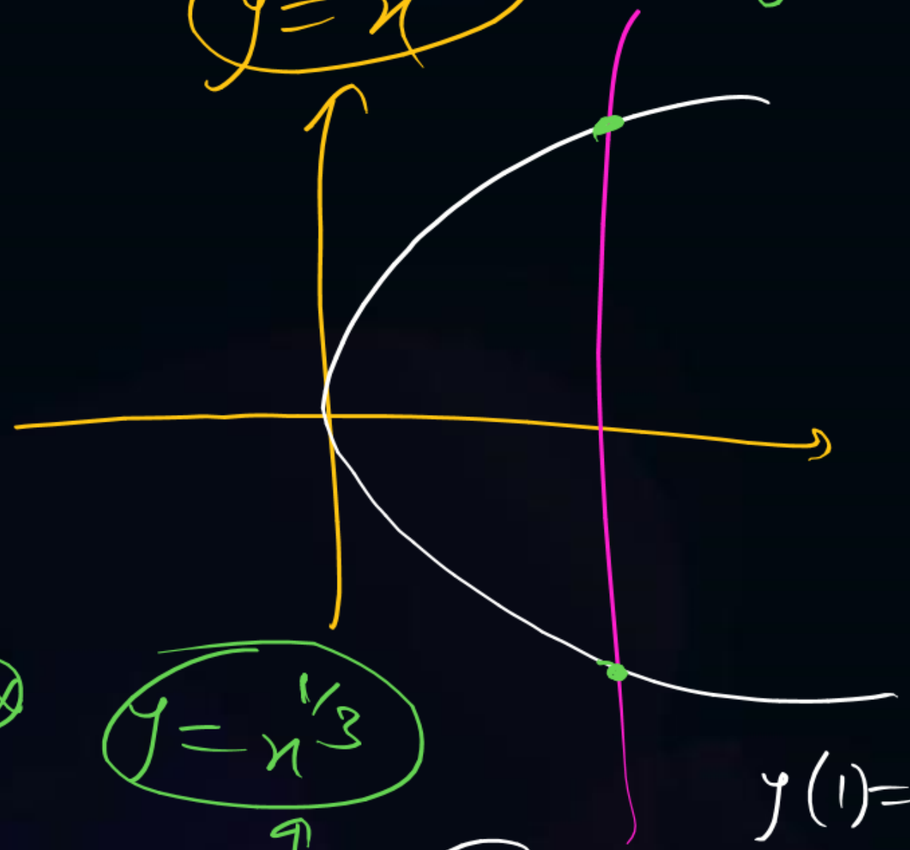
$D_{om} = (-\infty, \infty)$

$Range = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

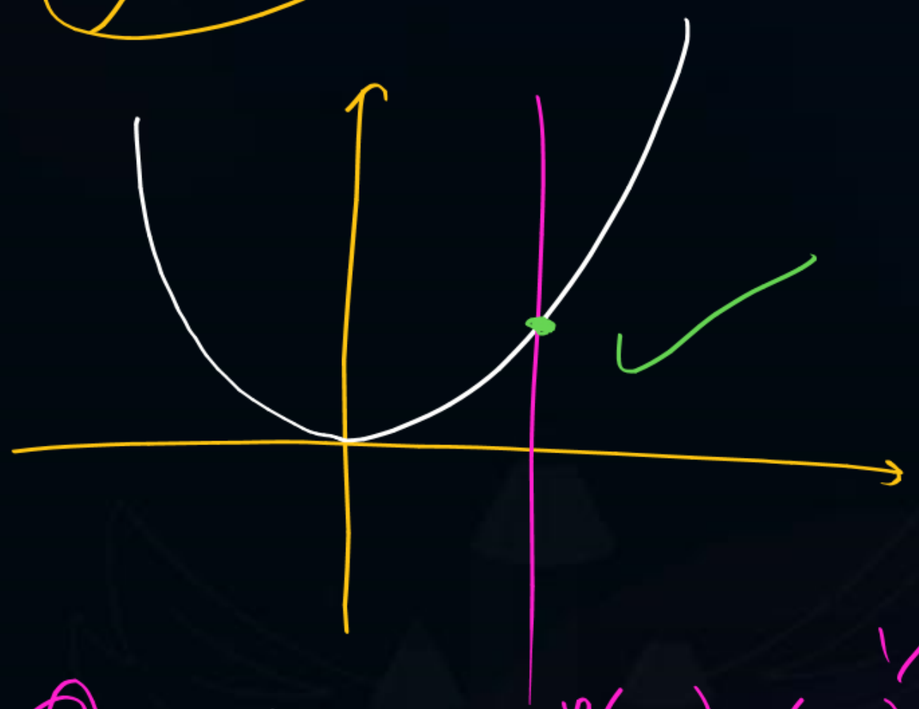
Not Periodic

odd func^{n}

(*) $y^2 = x$ ⇒ Not a func^n



$y = x^2$ ⇒ it is a func^n



(*) $y = x^{1/3}$



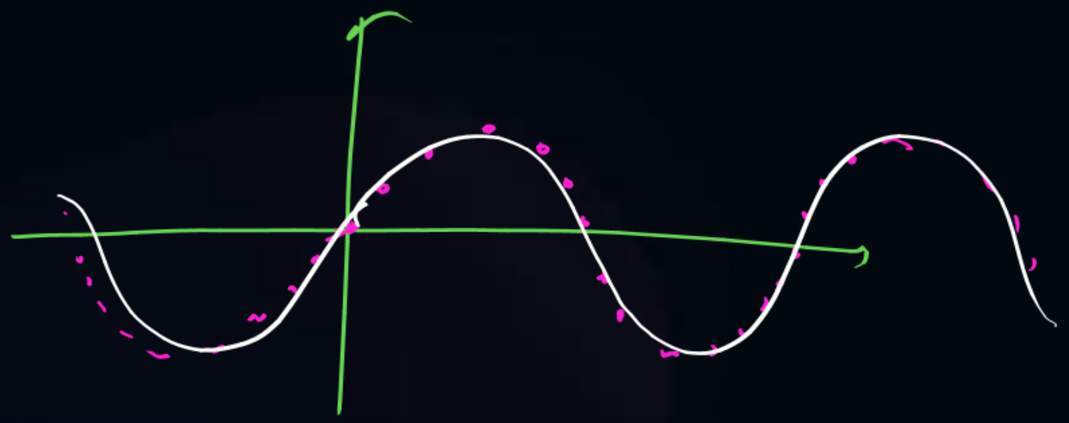
$y(1) = (1)^{1/3} = 1, w, w^2$

$y(-1) = (-1)^{1/3} = -1, -w, -w^2$

$y(-8) = (-8)^{1/3} = -2$

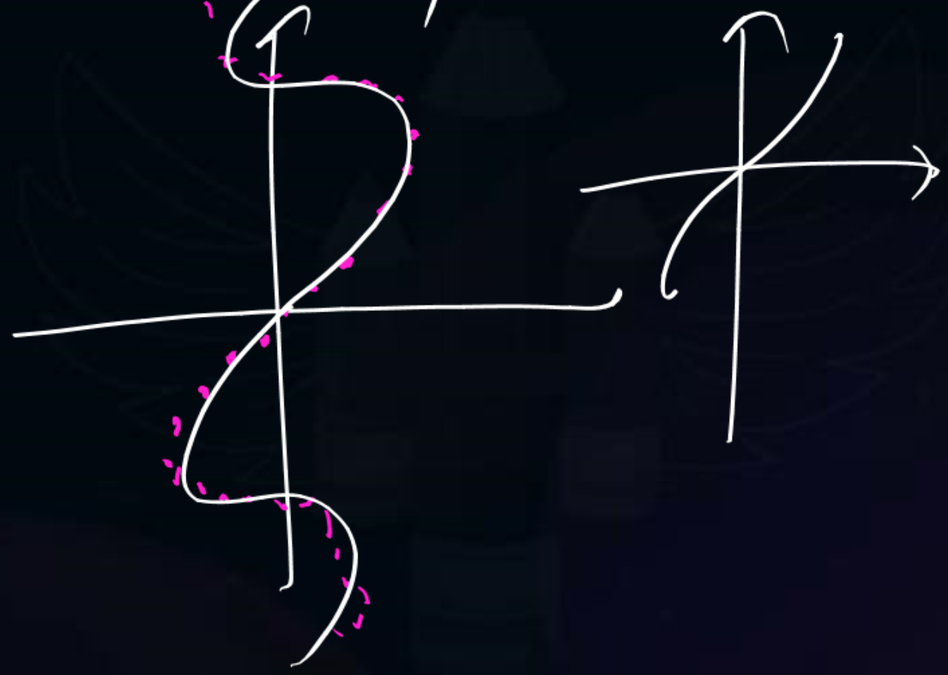
$y = \sin x$, $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{5\pi}{6}, \pi$ ---- —

$y = 0$, - - - - - - - - - - - - - - -



$y = \sin^{-1} x$, At $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$ - - -

$y =$ - - - - - - - -



$y = \tan x$, Domain $= R - \{\ldots -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots\}$

$$\text{Dom} = R - (2n+1)\frac{\pi}{2}$$ where $n = \text{Int}$.

for $n = -2$, $R - \left(\frac{-3\pi}{2}\right)$

, $n = -1$, $R - \left(\frac{-\pi}{2}\right)$

, $n = 0$, $R - \left(\frac{\pi}{2}\right)$

, $n = 1$, $R - \left(\frac{3\pi}{2}\right)$

, $n = 2$, $R - \left\{\frac{5\pi}{2}\right\}$

---

just skip it

if $n = \text{Int}$ then $(2n+1)$ is Representing odd Int

& $2n$ " " Even Int

THANK - YOU